

NightWatch

Hostel night attendance for BITS Pilani, Dubai Campus

Pitch and demo script

IEEE BPDC CampusOPS Hackathon · Problem statement PS03: Nighttime Attendance & Presence Scanning

Final round: 3 October 2026

Team: _____

How to use this document: Part 1 is what you say (about 5 minutes). Part 2 is the live demo, step by step (about 5 minutes), with a setup checklist and a fallback for every step. Part 3 prepares you for judges' questions. The appendix has logins and commands. Grey-shaded blocks are lines to say out loud; say them in your own words.

Part 1 · The pitch (about 5 minutes)

The one-line version

NightWatch replaces the 10:30 PM paper roll call with a 10-second phone check-in at the block reception, so the attendance taker only visits the rooms that actually need a visit, and the wardens get the same night's absences and a real curfew report instead of a notebook.

0:00 The problem today

Every night an attendance taker walks room to room with a paper list. Students who come back late write their room number in a notebook at reception. Nobody checks their name or the time.

Gate passes work like this: a parent emails the warden, the warden posts a screenshot in a staff WhatsApp group, and the guard scrolls through it. The pass is valid for one day only.

And right now, management is running a one-month pilot with an 11:00 PM curfew for girls. They said they'll reverse it if late coming continues, but today nobody has the numbers to show whether it's working.

0:45 What NightWatch does

NightWatch does three things. Students check themselves in on their phone. The taker only visits the rooms that didn't check in, plus a few random spot checks. And gate passes go from parent to warden to guard without WhatsApp screenshots.

It runs in any phone browser. There's no app to install.

1:15 How check-in works

There are two check-in times: 10:00 to 10:15 PM for students who sleep early, and 11:30 to midnight for everyone else. The student checks in once, in either, by scanning a QR code on the screen at their own block's reception. That code changes every 15 seconds, so a screenshot sent to a friend stops working almost immediately.

The phone also has to be on the campus Wi-Fi. The server checks the connection itself, so a fake-GPS app doesn't help, and a student who's out on mobile data can't check in.

After midnight they can still check in, but it's marked late. That late data is exactly what the curfew review needs.

2:00 The taker's round

Instead of knocking on every door, the taker gets a short list: only the students who didn't check in, plus about five random students who did. The random ones keep everyone honest, because nobody knows who'll be checked.

If the taker marks someone present by hand, they must choose a reason: asleep, sick, phone problem or other. If a spot-checked room is empty, the warden gets a high-priority alert that same night.

2:40 Gate passes

The student requests a pass in NightWatch. The parent gets an email with a one-time link and approves it in one tap, with no account needed. Then the warden approves it. The guard sees today's approved passes on their own page, and the student is automatically excused from attendance that day.

3:10 What the wardens get

Wardens see tonight's unexcused absences as they happen, not the next morning. They get a 14-night table of on time, late and absent for each block. And one click emails any student with three absences or five hand-marked check-ins in two weeks and asks them to meet the warden.

For the girls' curfew pilot, this table is the evidence management said they need.

3:40 Why it's hard to cheat

If a student tries to...	What stops it
Send a screenshot of the QR to a friend	The code changes every 15 seconds and is accepted for about 30 seconds at most.
Check in from outside campus	The server sees the phone is not on campus Wi-Fi and refuses the check-in.
Fake their location with a GPS app	NightWatch doesn't use GPS. The check is done by the server, not the phone.
Check in and then leave, or check in from another building	Random spot checks. An empty room raises a high-priority alert to the warden.
Always get marked by hand "because my phone died"	Every manual mark needs a reason; five in 14 nights flags the student to the warden.

4:10 What it doesn't do (say this, judges respect it)

We want to be clear about the limits. The Wi-Fi check proves a student is on campus, not that they're in their own block. A friend in the lobby could forward the live code to someone elsewhere on campus. The short code life and the random spot checks are our answer to that. If a block needs more, a small Wi-Fi router at its reception is a cheap upgrade.

Parent approval relies on the parent's email, which is exactly what the process relies on today.

4:35 What's next

Phase 2 has two parts. First, move NightWatch into the ERP as a Hostel section, next to Academics, HR and Faculty, with ERP login and the university's own student list. Only the ERP team can add a section, so today it runs beside the ERP. Second, connect the main-gate face scanner log to catch anyone who left without a pass. We haven't built either, because we don't have access yet.

4:55 Close

NightWatch turns a paper round and a notebook into a 10-second check-in, a shorter round for the taker, and real data for the wardens. Let us show you.

Part 2 · The live demo (about 5 minutes)

Setup checklist (do this 15 minutes before)

1. Laptop on **BITS-Student** Wi-Fi. Phone(s) on **BITS-Student**, battery charged, brightness up.
2. On the laptop: open a terminal in `nightwatch/backend` and run **python run_demo.py --pretend 23:35 --reset**. This loads fresh demo data and makes the server act as if it's 23:35, so check-in is open whatever the real time is. The pretend clock keeps ticking: you have about 25 minutes of on-time check-ins. If the demo slips, stop the server (Ctrl + C) and run **python run_demo.py --pretend 23:35** again, without **--reset**, which keeps the data.
3. Write down the phone address it prints next to "phones on BITS-Student use this" (for example <https://10.30.83.168:8000>). It changes if the laptop reconnects to Wi-Fi.
4. Turn on the laptop's **Mobile hotspot** (Windows settings). It's only used for the refusal in step 3.
5. If Windows Firewall asks about Python, click **Allow** (tick public networks too).
6. Laptop, **Chrome normal window**: go to <https://localhost:8000>, **guard.a@demo.edu**, open the reception screen and press Full screen. This is "the reception display".
7. Laptop, **Chrome private (incognito) window**: log in as **warden.a@demo.edu**. Laptop, **Edge**: log in as **taker.a@demo.edu**. (Each browser window type holds one login.)
8. Phone: open the phone address, tap **Advanced** → **Proceed** on the certificate warning, log in as **aarav@demo.edu**. Do this before the judges arrive so they don't see the warning.
9. Keep the terminal visible or one Alt+Tab away. Every check-in prints a [check-in] line there, and the parent's email appears there too.

All passwords: demo123. If the laptop's IP changes, the phone shows the certificate warning once more; tap Proceed again.

Step 1 The reception screen Laptop (reception display)

Do:

- Show the full-screen Block A QR code. Point at the bar under the code emptying.

Say:

This is the screen at Block A's reception. The code changes every 15 seconds, so a photo of it is useless half a minute later.

Expect: The code visibly changes when the bar runs out.

If it fails: If it says "Can't reach the server", check the terminal is still running and the laptop is still on Wi-Fi. It reconnects by itself within 3 seconds.

Step 2 A student checks in Phone (Aarav, on BITS-Student)

Do:

- Show the student page: "Not checked in", with both check-in times (22:00–22:15 or 23:30–00:00).
- Tap **Scan QR at reception**, point the phone at the laptop screen.

Say:

Aarav is on the campus Wi-Fi. He scans the code at his block's reception, and he's done. That took about ten seconds.

Expect: Green "Marked present". The terminal prints a [check-in] line: Aarav, his phone's address, present.

If it fails: If the camera doesn't open, check Safari/Chrome camera permission for the site. If the scan says "QR code expired", just scan again (the code had just changed).

Step 3 Someone not on campus Wi-Fi is refused Phone on the laptop hotspot (Vivaan)

Do:

- Switch the phone's Wi-Fi to the **laptop's hotspot**. Open **<https://192.168.137.1:8000>** (Proceed on the warning if shown), log in as **vivaan@demo.edu**.
- Scan the same reception screen.

Say:

Now Vivaan's phone is on a network that isn't the campus Wi-Fi. In real life that's mobile data. For the demo we use the laptop's hotspot, because a phone on mobile data can't reach a laptop on campus at all. Same code, same block, and the server refuses him.

Expect: Red "Connect to the campus Wi-Fi (BITS-Student), not mobile data". The terminal prints REFUSED, not on campus Wi-Fi.

If it fails: A second phone makes this step faster: keep phone 2 on the hotspot from the start. Afterwards, switch phone 1 back to BITS-Student.

Step 4 Gate pass: student, parent, warden Phone (student), terminal, laptop (warden)

Do:

- On the phone (back on BITS-Student), log out and log in as **ishaan@demo.edu**. Under Request a gate pass, keep today's date, type a reason, tap **Send request**.
- In the terminal, show the email to the parent. Copy the **<https://.../parent.html?t=...>** link and open it (laptop or phone). Tap **Approve**.
- In the warden window, press Refresh: the pass is under "Gate passes waiting for you". Click **Approve**.
- On Ishaan's phone, reload: My passes shows **Approved**.

Say:

Ishaan asks for a pass. His parent gets a one-time link by email and approves with one tap, with no account needed. Then the warden approves. The guard sees it on his page, and Ishaan is excused from tonight's attendance. No WhatsApp screenshots.

Expect: Parent page says "Thank you. You approved the request." Opening the same link again says it's already used.

If it fails: In the demo, emails are printed in the terminal instead of being sent. Say so if asked: switching to real email is a configuration change.

Step 5 The taker's round Laptop (Edge, taker)

Do:

- Show "Rooms to visit": only students who haven't checked in. Ishaan isn't on the list, because he's excused.
- For one student, pick **Asleep** and tap **Present**. For another, tap **Absent** and confirm.

Say:

The taker doesn't walk every corridor. They see only who's missing. Marking present by hand needs a reason, so we can spot the student whose phone is always "dead". Marking absent alerts the warden right away.

Expect: The marked students disappear from the list. Tonight's counts update.

If it fails: Spot checks appear once students have checked in. With only Aarav checked in, he may show up as a spot check. That's fine, and worth pointing out.

Step 6 The warden's view Laptop (private window, warden)

Do:

- Press Refresh. "Absent tonight, no gate pass" shows the student the taker just marked absent.
- Click the arrow on **Last 14 nights** to open the table: one row per night with on time, late, absent and marked by taker.
- Click **Run the 14-night check**. Rohan is flagged; show his email in the terminal.

Say:

The warden sees tonight's absences tonight. This table is two weeks of on time, late and absent, the evidence management needs for the curfew pilot. And one click emails every student who crossed the limit.

Expect: Rohan, Room A105, flagged with his absences; an email line in the terminal.

If it fails: The table uses demo history created by --reset. Say so if asked: "these past nights are sample data; tonight is live."

Close the demo: "That's a full night: check-in, a refusal, a gate pass, the taker's round and the warden's report. Happy to take questions."

Part 3 · Judges' questions

Short, honest answers. If you don't know something, say what you'd need to find out.

Q: What if a student sends the QR code to a friend who isn't in the hostel?

A: The friend must be on campus Wi-Fi and use it within about 30 seconds. If they're off campus, it's refused. If they're elsewhere on campus, it can work, which is why spot checks exist: that student's room may be checked, and an empty room is a high-priority alert.

Q: Why not use GPS?

A: GPS is easy to fake with free apps and unreliable indoors. The Wi-Fi check happens on the server, and the phone can't lie about which network its connection came from.

Q: What if a student's phone is dead or they don't have one?

A: The taker marks them present by hand with the reason "phone issue". That's the same as today for that one student. Five manual marks in 14 nights flags it to the warden.

Q: What if the Wi-Fi or the server goes down?

A: The taker does the full round on paper, which is exactly what happens today. NightWatch removes work when it's up; it doesn't add risk when it's down.

Q: Does this track students' location?

A: No. It stores the time of the check-in and the network address of that one request, nothing else. No GPS, no background tracking.

Q: Why isn't it inside the ERP, like Academics, HR and Faculty?

A: That's where it should end up, as phase 2. Only the university's ERP team can add a section to the ERP, and we needed something we could build and show working for this round. We've already matched the ERP's look so it feels familiar. Moving in means students log in with their ERP account and the student and room list comes from the university instead of our own copy.

Q: Would the Wi-Fi check still work from inside the ERP?

A: Mostly. The ERP runs on Oracle Cloud, outside campus. From there, every phone on campus Wi-Fi shows up with the campus's shared internet address, not its own campus address. That still tells campus Wi-Fi apart from mobile data, which is the main check. We'd lose the detail of which part of the campus network the phone is on, and the campus address list would need updating if it ever changes. The rotating QR at the reception still proves the student is at their block. The other option is to keep the check-in step on a small campus server and put everything else in the ERP.

Q: What about the main-gate face scanner?

A: Also phase 2. Its exit and entry log would show anyone who left without a pass. We didn't integrate it because we have no access to its data.

Q: What does it cost to run?

A: One small server on the campus network and a screen at each of the six block receptions. An old monitor or a basic tablet is enough. No app store fees; it runs in the phone's browser.

Q: How are accounts and passwords protected?

A: Passwords are stored hashed (bcrypt), logins use signed tokens that expire after 12 hours, every page checks the user's role and block, and parent links work once and expire after 24 hours.

Q: Why did you leave out complaints?

A: The university's ERP support portal already handles complaints, and the Student Council asked students to use it. We didn't want to build a second channel.

Q: Is the 14-night data real?

A: The past nights in the demo are sample data we generated. Tonight's check-ins in the demo are live.

Appendix · Logins and commands

Demo logins (password for all: demo123)

Role	Email
Chief warden (all blocks)	chief@demo.edu
Block A warden / taker / guard	warden.a@demo.edu · taker.a@demo.edu · guard.a@demo.edu
Block G warden / taker / guard	warden.g@demo.edu · taker.g@demo.edu · guard.g@demo.edu
Block A students	aarav, vivaan, ishaan, kabir, reyansh, arjun, dhruv, vihaan, rohan, karan (@demo.edu)
Block G students	ananya, diya, saanvi, myra, aadhya, kiara, riya, meera, tara, nisha (@demo.edu)

Everyone uses the same login page; it opens the right page for each account.

Commands (run in nightwatch/backend)

Command	What it does
python run_demo.py --pretend 23:35 --reset	The demo: fresh data, server clock set to 23:35 (about 25 minutes of on-time check-ins), HTTPS, campus Wi-Fi check on.
python run_demo.py --pretend 23:35	Restart the pretend clock but keep tonight's data.
python run_demo.py	Same server with real time (for a demo actually held at night).
python run_demo.py --no-wifi-check	Rehearsal off campus: the Wi-Fi check is off.
python seed.py then python test_flow.py	Automatic tests (15 checks). Run seed.py again afterwards.
Ctrl + C in the terminal	Stops the server.

Addresses

- Laptop: **https://localhost:8000**
- Phones on BITS-Student: the **10.30.x.x** address printed by run_demo.py
- Phones on the laptop hotspot: **https://192.168.137.1:8000** (check-in is refused here, on purpose)

Facts to have ready

- 6 hostel blocks (4 boys, 2 girls); the demo data has 2 of them (Block A, Block G).
- Check-in 22:00–22:15 or 23:30–00:00, once a night; after 00:00 it's allowed but marked late. A "night" runs noon to noon.
- Main gate curfew: 11:30 PM boys; 11:00 PM girls (one-month pilot). Girls' hostel in-time: 12:00 midnight.
- Flag rule: 3 absences or 5 manual marks in any 14 nights leads to an email and a meeting with the warden.